

How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound

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Abstract

Unilateral contact between reduced (modal) deformable bodies and rigid bodies is commonly resolved by a velocity-level complementarity law in which one shared multiplier couples the rigid and modal unknowns. Interactive solvers evaluate this law under small, fixed local iteration budgets; we measure the energetic cost of that truncation. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene $\times$ relaxation $\times$ budget sweep, we observe sharply formulation-dependent behaviour. Measured against the gross rigid-side loss (a scene-wide, gravity-corrected quantity, not a per-contact dissipation), the position-based host overdraws by up to $4.4 \times 10^7 \text{ J} - 1.2 \times 10^5$ times the incident rigid kinetic energy in the most starved cells — whereas the augmented-Lagrangian host overdraws in 23 of 24 cells by at most 15 J, and the implicit realization never overdraws. These are distinct pathologies — one pervasive but negligible, the other rare but catastrophic — and the customary ratio-to-incident-energy diagnostic does not separate them. An iteration-budget sweep identifies the position-based amplification as a truncation artifact: it decays monotonically toward a converged reference computed in the same code path, closing the gap by six orders of magnitude. Since the converged regime is unaffordable at interactive budgets, we introduce a cumulative modal-storage bound funded by the measured gross rigid-side loss and enforced by a reservoir ledger and a projection of the realized modal state. The bound holds in all 90 measured cells (78 distinct configurations), including the 1×8 and 2×4 budgets at which interactive solvers are commonly deployed. Enforcement in turn costs contact validity: worst-case post-projection penetration reaches 21.6 mm, 72% of the supporting board's thickness.

CCS Concepts

• Computing methodologies $\rightarrow$ Physical simulation.

Keywords

contact, modal reduction, passivity, position-based dynamics

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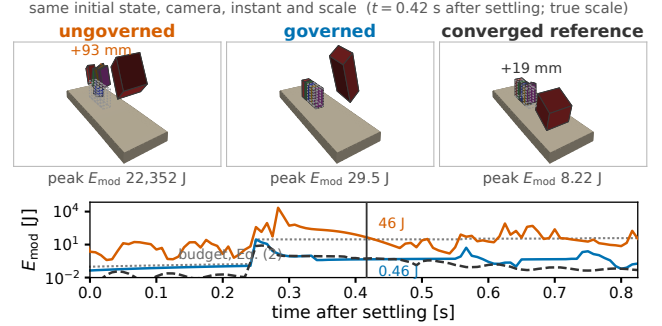


Figure 1: One shelf impact at a production-like interactive budget (one iteration per substep, eight substeps per frame), true scale, three independent runs. Ungoverned, truncation energy throws the resting books up to 93 mm — more than their own height; governed, (2) holds peak modal energy to 29.5 J and they stay within 1 mm of rest; the reference is the host's own converged solution ($K=500$). The bound removes the spurious launch and some legitimate motion with it: the reference lifts one book 19 mm where the governed run lifts none (§3.3). The impactor's own rebound is rigid-side, which the bound does not govern (§4).

1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with rigid bodies, the established treatment resolves rigid and modal unknowns together through a shared contact multiplier at the velocity level [13]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [1, 6]. Interactive modal deformation subject to collision and contact constraints predates both [5].

We take this coupled row as given. Written at position level it sits one linearization step from the velocity-level law: for a gap function C , $C^{n+1} \approx C^n + h\mathbf{J}\mathbf{u}^{n+1}$, which divided by h recovers the velocity-level complementarity condition with a C^n/h stabilization bias, carrying the same Jacobian and the same multiplier as Zheng and James [13]. The row itself is *not* a contribution of this paper, and our transcription is narrower than the original law: modes couple through normal rows only, and the position-level form assumes $e = 0$ restitution.

Interactive solvers, however, do not solve this row to convergence: they run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition

boundary can inject spurious energy even when every subsystem is individually passive [11] — an observation that likewise predates this work. Velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [7]. What is missing is a quantitative account of this case — unilateral contact-to-modal transfer — and a mitigation that fits within an interactive budget.

Contributions.

- (1) **Measurement.** A single contact row evaluated in three fixed-budget solver formulations over one identical 24-cell adversarial sweep on one machine (§3.1), plus an iteration-budget sweep against a converged reference identifying the position-based injection as a truncation artifact (§3.2).
- (2) **A cumulative storage bound.** A one-directional bound on the energy entering modal storage, funded by the measured gross rigid-side loss and enforced by a reservoir ledger and a projection of the realized modal state (§2); it satisfies the prescribed inequality in all 90 measured cells (78 distinct configurations).
- (3) **The cost of enforcement.** Post-projection contact validity (penetration, corrective impulse, multiplier variance, momentum), measured rather than assumed (§3.3).

Relation to concurrent energy-control work. Wei et al. [11] guarantee discrete passivity for any finite inner-iteration budget via Douglas–Rachford splitting in wave coordinates whose Fejér monotonicity supplies algorithmic dissipation; their interfaces are *bilateral* couplings, not unilateral contact, and the guarantee is structural rather than a budgeted transfer cap. You et al. [12] modify implicit Euler to track user-specified *total-energy* targets bidirectionally on full nonlinear elastodynamics with barrier contact. Ours is neither: a one-directional cap on what contact may deposit into modal storage, funded by measured gross rigid-side loss and enforced by scaling realized modal state. For flexible bodies in a game engine, Zobel et al. [14] solve ray-traced penalty contact quasi-statically without a shared unilateral multiplier; modes and modal derivatives have been used in a position-based host as a *deformation* surrogate [9], whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier.

Relation to passivity control and energy projection. The governor transplants an established control idea rather than inventing one. Time-domain passivity control [4] damps once an energy observer goes negative; energy-tank formulations [3] give that observer explicit state, so a controller spends only what its tank holds; Rath [10] enforces exact per-interface energy stability for contacting modal objects at audio rate, where we cap a scene-level scalar reservoir. We keep that structure and change two things: the tank is funded by *measured gross rigid-side loss* — energy the solve removed elsewhere — rather than by the port it regulates, and the actuator scales realized *state* rather than modulating a force, because a fixed-budget solver commits its multipliers before the excess is observable. FEPR [2] also projects state post-solve, but onto conservation of *total energy*; ours is one-sided and never restores energy the solver lost.

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law.

The invariant. We require that the energy stored in the modal subsystem never exceed the *measured gross rigid-side loss* — defined exactly at (3) below — cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1]. \quad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs $\eta = 1$* , the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total remains within the budget, which is what lets a legitimate ring persist across substeps.

Definitions. Modal storage is the mechanical energy of the reduced state as written above, and $E_{\text{mod}}^0 = 0$: every scene starts from an undeformed, resting support, so even the run’s own settling transient must be funded by the reservoir. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g}(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \omega_b^\top \mathbf{I}_b \omega_b]$ is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and $\pm$ denote the two ends of substep k . We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid–rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§4). Gravity enters only through the work term W_g , so free ballistic motion registers zero supply: a falling body’s kinetic gain exactly equals the gravity work, and only an inelastic velocity change funds the reservoir. Substep-boundary endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

What is guaranteed, and what is not. Equation (2) is a *cumulative, gross-loss-funded storage ceiling on the modal subsystem*, with a supply that is *measured, not prescribed*: the right-hand side is the measured loss of (3), never a tuned constant or a fraction of incident energy. It is not contact-port passivity and not a signed per-interface transfer bound — it constrains a scalar reservoir, not a port. The *gross* sum ignores substeps in which the rigid subsystem *gains* energy, so those cannot cancel earlier credits, and energy returned and re-dissipated counts as fresh supply; §4 measures both.

The enforced inequality is slightly weaker than the stated one. Reported violations use (2) exactly as printed. The implementation’s own test additionally forgives one substep’s largest deposit, $\max_k \eta \max(\Delta E_{\text{rig}}^k, 0)$, absorbing a granularity artifact of the loop

below: modal potential energy can rise in the very substep whose credit funds it. That allowance is worth 7–389 J here — immaterial where the overdraw is 10^5 J or more, decisive where it is not: under it the augmented-Lagrangian host would violate 1 of 24 cells rather than 23. We report the stricter reading throughout, for governed and ungoverned alike; the governed runs satisfy it with a worst margin of 1.1×10^{-13} J.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, within the same substep: ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. If the realized modal energy would overdraw B , the state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma (\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma = \min\left(1, \sqrt{\frac{E_{\text{mod}}^- + B}{E_{\text{mod}}^+}}\right), \quad (4)$$

with $E_{\text{mod}}^\pm$ the modal energy before and after the solve. One substep of the governed loop, identical in all three hosts, then reads:

- (1) snapshot E_{rig}^- (pure KE, all bodies), E_{mod}^- , $\mathbf{x}^-$;
- (2) predict (gravity); generate contacts at the predicted pose;
- (3) position solve (K iterations), then velocity solve — *the contact solve*;
- (4) $\Delta E_{\text{rig}} = (E_{\text{rig}}^- - E_{\text{rig}}^+) + W_g$, i.e. (3), post-solve;
- (5) credit $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ — same substep;
- (6) if $E_{\text{mod}}^+ > E_{\text{mod}}^- + B$, scale the state by γ of (4);
- (7) debit $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. Contact is **not** re-solved.

The form (4) is closed because E_{mod} is quadratic in the state: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both the kinetic and the elastic term by γ^2 , so γ bounds the total energy rather than the velocity alone. Two consequences are measured below: the projection is inert whenever the solve is within budget ($\gamma = 1$, bitwise-identical trajectories), and it never re-solves contact (step 7).

3 Results

All solver-behaviour measurements below come from a single machine (Table 1), the sole exception being the device-resident timings of §3.5 (NVIDIA RTX 4090), reported separately; every number is frozen with its generating command and commit hash in the supplemental ledger. We deliberately avoid mixing machines: chaotic contact stacks diverge across architectures under floating-point reassociation.

3.1 One row, three formulations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes — a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [1] — and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. Table 1 lists what the hosts do differently and what is pinned identical. Two entries deserve emphasis. The relaxation axis is *inert* on the impulse host: its implicit modal weight needs no under-relaxation and the attribute is never read (hence the bit-identical hatched rows of Fig. 2). Warm-start policy also differs. We run each host as published: $\lambda \leftarrow 0$ every substep is the position-based host’s own form, and carried duals with a growing

Table 1: Host differences (above the rule) and settings held identical (below it). Entries are read from the implementation, with source anchors in the supplemental ledger.

	position-based	augmented-Lagr.	seq. impulse
unknown solved for	position	position + multiplier	velocity
contact treatment	compliant projection, $\hat{\alpha} = \alpha/h^2$ ($\alpha=0$: rigid)	penalty + dual update, $\alpha=0.99$, $\beta=10^5$	Schur tem, $\mathbf{A} = \frac{\mathbf{c}_{\text{fin}}}{h^2} \mathbf{I} + \mathbf{J}\mathbf{M}^{-1}\mathbf{J}^\top + \mathbf{G}\mathbf{W}\mathbf{G}^\top$
modal weight	explicit; per-mode compliance $1/(H_{ii}h^2 - 1)$	block-GS, full Newton step on $\mathbf{q}$	<i>implicit</i> , $\mathbf{W} = (\mathbf{M} + h\mathbf{D} + h^2\mathbf{K})^{-1}$
iteration	GS sweeps, gap re-evaluated each sweep	coloured primal GS + $\mathbf{q}$ -block	PGS over rows
warm start	none ($\lambda \leftarrow 0$ each substep)	λ and penalty carried over	λ cache keyed per row
relaxation	0.7 on $\mathbf{q}$ and the support block	0.7 on the $\mathbf{q}$ -block Newton step	<i>inert</i> : never read
h ; substep	1/120 s; h/S , contacts regenerated per substep		
modal rank r	24 (shelf), 28 (ledge), 24 (table)		
modal damping stepper; η	Rayleigh $\mathbf{D} = \alpha_0\mathbf{M} + \alpha_1\mathbf{K}$, $\alpha_0 \in \{2, 3\}$, $\alpha_1=10^{-5}$		
machine	implicit midpoint; $\eta = 1$ Apple M4, CPU only, CPython 3.12		

penalty are constitutive of the augmented-Lagrangian method, so neither is our choice; the impulse host’s per-row λ cache is the only genuine policy default among the three. The comparison is therefore as-deployed, not compliance-matched.

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis spans $\times 64$ and conflates two refinements that are neither interchangeable nor equally priced — a substep also repeats contact generation and integration, so equal-sweep cells are not equal-cost. With a fixed budget of 32 row evaluations on the shelf drop, spent as iterations ($K=32$, $S=1$) the ratio is 0.300 and (2) holds; spent as substeps ($K=4$, $S=8$) it is 3.13, overdrawing by 481 J: substep refinement alone does not converge the row (the full ladder is in the supplement).

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1 : peak modal energy 4.44×10^7 J against an incident 371 J). The augmented-Lagrangian host exceeds $R = 1$ in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is not unconditionally benign either. The ordering is scene-dependent, and it inverts: taking the worst over scenes at 4×1 , the position-based host is five orders of magnitude more energetic (1.2×10^5 vs 1.70), yet on the table scene itself the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. An impactor’s incident kinetic energy is not the supply (2) budgets against, so we measure (2) directly on all three hosts, ungoverned, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an ungoverned run (the ratios above reproduce the independent sweep exactly).

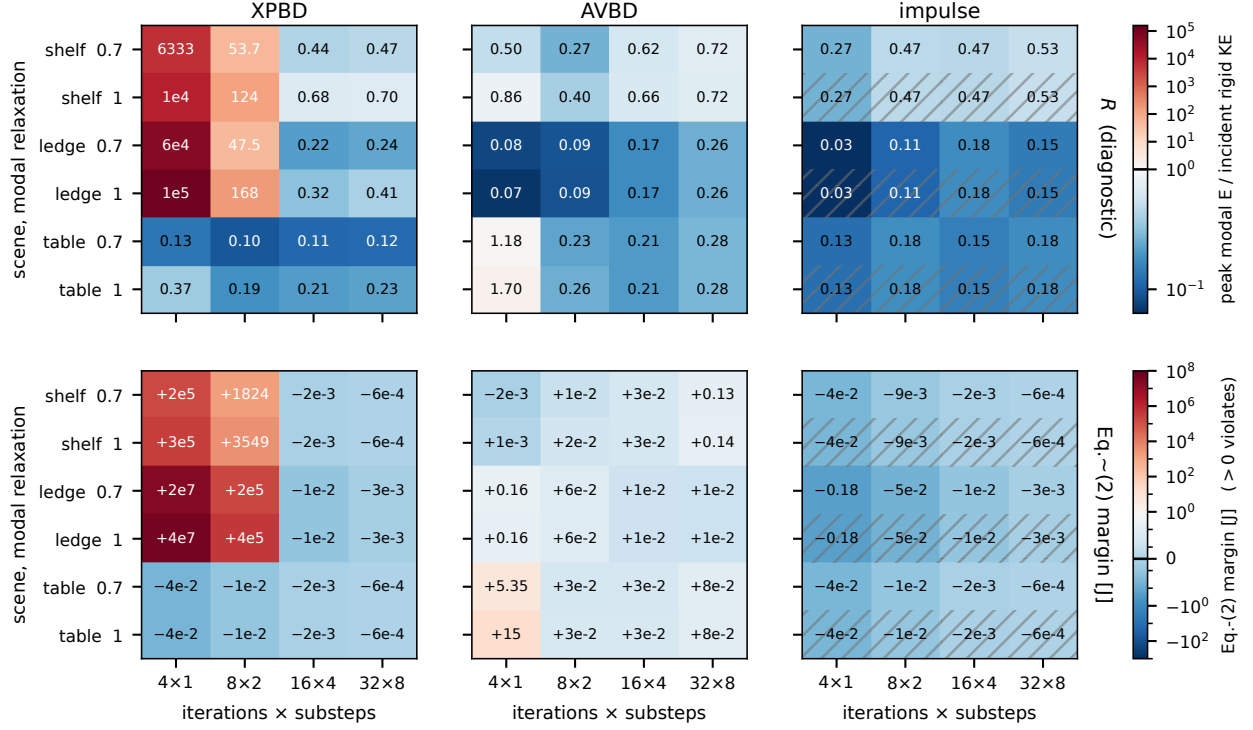


Figure 2: The same contact row in three fixed-budget solver formulations over an identical 24-cell sweep (3 scenes $\times$ modal relaxation $\{0.7, 1.0\} \times$ budget), governor OFF. Top: the ratio $R = \text{peak modal energy} / \text{peak incident rigid kinetic energy}$, log scale diverging at 1 – a severity diagnostic. XPBD exceeds 1 in 8/24 cells, AVBD in 2/24, the implicit realization in 0/24. Bottom: the invariant itself – the signed margin of (2) in joules, symmetric-log, diverging at 0; positive values violate. The two panels disagree on AVBD. Hatched rows mark the relaxation axis where the formulation does not consume it; those rows are bit-identical by construction, shown only for grid symmetry. We report joules rather than a second ratio because every candidate denominator misleads: the supply-to-date explodes in the opening substeps, and the run total hides early peaks.

We report the signed margin of (2) in joules; positive values violate. The two metrics agree on the position-based host (8/24 either way) and on the implicit one (0/24, margin at worst -5.7×10^{-4} J), but they disagree on the augmented-Lagrangian host, which violates (2) in 23 of 24 cells although R flags only 2. The magnitudes separate the two failures: in 21 of its 23 cells the augmented-Lagrangian overdraw is below 0.15 J (worst 15.1 J), against the position-based host's 4.4×10^7 J. A pervasive but negligible overdraw and a rare but catastrophic one are different pathologies, and only the invariant distinguishes them.

With enforcement enabled, all 72 cells of this sweep satisfy (2) – 60 of them distinct configurations, the implicit host's 12 relaxation pairs being bit-identical (Fig. 2); the 18 deployed-budget cells of §3.2 satisfy it too. The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell, nor at the well-resolved budgets (16x4, 32x8) on the position-based host at relaxation 0.7: it is inert precisely where the solve is already within budget.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the converged reference 0.2735; the gap $|\text{XPBD} - \text{reference}|$ shrinks from 2.96×10^4 to 2.7×10^{-2} . The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (spread 4.9×10^{-4}).

A falling energy ratio does not by itself show that the row is converging – a single scalar could shrink coincidentally. Measuring the constraint directly resolves the question: the end-of-substep complementarity residual $\|\min(C, \lambda)\|_\infty$ on the same sweep falls monotonically from 2.79×10^{-2} at $K=1$ to 3.56×10^{-6} by $K=64$, where it saturates ($K=128$ moves it only in the seventh significant figure). The energy crossing at $K \approx 24$ coincides with the residual passing $\sim 3 \times 10^{-3}$; the amplification disappears precisely as the complementarity conditions begin to hold. We report this for the position-based host only: the augmented-Lagrangian multiplier is not the same object, and the implicit realization is already converged at $K=2$, so neither yields a comparable curve.

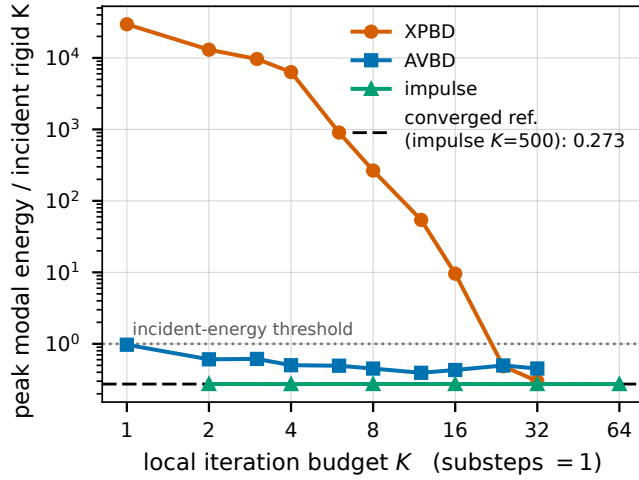


Figure 3: Energy ratio versus local iteration budget K on a deterministic shelf drop, substeps pinned at 1 so K is the only variable, governor OFF. The position-based host decays monotonically toward the converged reference, crossing the incident-energy threshold between $K=16$ and $K=24$; the gap shrinks by six orders of magnitude. The reference is the implicit realization at $K=500$ — the same code path run to convergence, not a different model; it is already converged at $K=2$. The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small K , so it cannot be read as a convergence trend.

Self-convergence stops short of the reference. Running the position-based host to $K=500$ against itself does not close the remaining gap: its ratio plateaus by $K \approx 64$ (0.3003, 0.2999, 0.2997, 0.2996, 0.2996 at $K=32 \dots 500$), so the residual 2.6×10^{-2} against the reference’s 0.2735 is a fixed point, not a tail. The state agrees no better: on the deflection trajectory $d_i(t) = U_y[i]^T \mathbf{q}(t)$ the peak matches to 2.4% but the trajectories differ by $\|d^{500} - d^{\text{ref}}\|_\infty = 6.8$ mm, 34% of the reference’s own peak, likewise flat from $K=32$. The two hosts converge to different discrete solutions — expected, since they discretize the same continuous law differently — so the sweep separates two easily conflated effects: a truncation pathology that convergence removes, superimposed on a formulation difference that it does not. (No ring frequency is reported here: quasi-static sag dominates these windows, so the dominant spectral peak moves with window length; §3.4 has the resolved comparison.)

This central diagnostic cuts in two directions: the amplification is created by truncation and largely cured by convergence, so iterating further — or adopting the implicit formulation — would remove the pathology outright. The bound of §2 is therefore not a substitute for convergence; it targets the regime where neither escape is available. Following the one-iteration small-step regime [8], inter-active position-based solvers run schedules near 1×8 to 2×4 , well below the $K \geq 24$ this scene needs; these are representative stress budgets, not universal engine defaults. Measured on 18 further cells (3 scenes $\times$ 3 hosts $\times$ $\{1 \times 8, 2 \times 4\}$, relaxation 0.7), the position-based host violates (2) in all six of its deployed cells, worst $R = 2282$ — on 8 row evaluations, twice the 4 of the 4×1 corner. The other

Table 2: Post-projection contact validity at starved budgets, per clamp-active substep. Gap violation is measured immediately after the γ -scale; the corrective impulse is the next substep’s normal impulse against the unclamped steady-state median, a ratio because the hosts’ multipliers are different objects. The 4×1 position-based cells clamp on nearly every substep, so no steady state exists there (—). The augmented-Lagrangian rows are its only two cells with $R > 1$.

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
<i>position-based, relaxation 0.7</i>					
shelf	4×1	107/108	1.25 / 21.6	—	—
shelf	8×2	162/216	0.30 / 6.46	$6.90 \times$	$26 \times$
ledge	4×1	98/108	0.07 / 21.2	$1.03 \times$	$1.07 \times$
ledge	8×2	96/216	0.07 / 4.08	$8.69 \times$	$58 \times$
<i>augmented-Lagrangian</i>					
table (0.7)	4×1	27/108	0.01 / 1.43	$1.17 \times$	$0.41 \times$
table (1.0)	4×1	25/108	0.02 / 3.11	$1.39 \times$	$1.38 \times$

hosts behave as in the sweep (5/6 and 0/6 cells, the former by at most 0.16 J). Governed, all 18 satisfy (2) (worst $R = 1.022$), and the projection is gentler than at 4×1 : worst penetration 9.8 mm against 21.6.

3.3 The cost of enforcement

Accuracy. Bounding the energy is useful only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8×2 , relaxation 0.7), peak modal energy is 1555.6 J ungoverned and 29.26 J governed against the converged reference’s 7.92 J: the energy error falls $196 \times \rightarrow 3.7 \times$, or $+1548 \rightarrow +21$ J; against the host’s own fixed point (8.22 J), $189 \times \rightarrow 3.6 \times$, so the result does not depend on the reference chosen. The trajectory, however, degrades: $\|d - d^{\text{ref}}\|_\infty$ is 6.5 mm ungoverned and 14.3 mm governed, 33% and 71% of the reference’s peak — the governor acts as a safety envelope, not an accuracy device. At the deployed 1×8 cell the same comparison gives $2823 \times \rightarrow 3.7 \times$ on energy and the trajectory improves as well ($108 \rightarrow 96\%$ of reference peak). The trade is budget-dependent, but 96% of the reference peak is still far from faithful.

This also explains how a $196 \times$ energy error can coexist with a peak deflection only $1.2 \times$ too large. The excess is neither kinetic (0.3%) nor low-frequency: this scene’s spectrum is bimodal (ten bending modes below 2.1 kHz, six stiff ones above 20 kHz), and 99.6% of the ungoverned energy sits in the stiff cluster, $\sim 200 \times$ above the 240 Hz substep rate, against 0.0% for both references. Being a single scalar, γ cannot redistribute energy across the spectrum: it corrects the *amount* ($1555.6 \rightarrow 29.3$ J), leaves the character intact ($99.6 \rightarrow 97.7\%$), and scales the legitimate sag down with the noise ($24.1 \rightarrow 5.9$ mm against 20.0).

Contact validity. Since $\gamma < 1$ shrinks the sag $U_y^T \mathbf{q}$, (4) lifts the support *into* the resting body without re-solving contact. Table 2 quantifies the consequence, instrumented by wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.25 mm, but the worst case reaches 21.6 mm, a $3.5\text{--}12.6 \times$ increase over its pre-scale value. The shelf board is 30 mm thick, so that is 72% of its thickness;

against the unclamped scene at a converged budget (resting sag 1.7–2.4 mm, peak dynamic deflection 20.0–20.5 mm) it is 9–13× the sag and 1.05–1.08× the board’s peak deflection — a visible interpenetration, not a sub-millimetre artifact, occurring where the projection is most load-bearing. Contact recovers in the next substep rather than accumulating, but that correction costs a normal impulse of up to 8.7× the steady-state value, with multiplier variance up to 58× higher. Net rigid linear momentum is unchanged across the projection — with exactly zero drift — in every clamp-active substep of every cell, necessarily so since (4) writes only modal state; we verified this numerically rather than asserting it.

The same instrumentation on the augmented-Lagrangian host, at its only two cells with $R > 1$, finds the projection an order of magnitude gentler: a quarter of substeps rather than nearly all, γ no lower than 0.60, worst-case violations of 1.4 and 3.1 mm against 21.6, and corrective impulse and multiplier variance within 1.4× of steady state where the position-based host reaches 8.7× and 58×. This is consistent with §3.1 — its overdrafts are at most 15 J — while the *relative* effect is comparable (3.1 and 5.5× against 3.5–12.6×). The hosts differ in the scale of the correction rather than in its character.

3.4 The reduced response against a full-FEM reference

Bounding the energy is likewise useful only if the bounded response is the correct one. On the ledge scene we compare the reduced arm against an unreduced FEM solve of the same mesh, both built on the *same* discrete operator (the reduced basis is $\text{eigsh}(\mathbf{K}, \mathbf{M})$ of the operator the reference integrates), so discretization bias cancels from every ratio. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as h refines $1/120 \rightarrow 1/960$, plateauing near 0.9; the residual $\sim 10\%$ is $k=24$ mode truncation on a thick, high-frequency slab, not a coupling error — the ring frequency already agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees too: peak deflection sits at the pedestal antinode rather than under the impact, and the falloff tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term. This is the pre-topple, small-displacement regime; we claim nothing beyond it.

3.5 Runtime cost

On the CPU host at 16×4, over ten repetitions of 100 frames, the baseline step costs 11.0–125.6 ms by scene and formulation, and the ledger adds 0.23–0.41 ms — 0.9–3.4% — on the four shelf and ledge configurations, where it resolves above run-to-run variance. On the table scene it does not: the overhead’s per-repetition spread exceeds its mean in both hosts ($+1.35 \pm 2.08$ and $+0.04 \pm 1.43$ ms), so we report it only as bounded by ~ 1.4 ms. The percentage is the portable quantity: on a second machine absolute times differ by 3.5–4.6× while the percentages agree to within 0.02 points wherever both resolve. These baselines fall 1.3–15× short of a 120 Hz budget, though shelf and ledge meet it at 2×4 with the governor on.

A separate device-resident path (NVIDIA RTX 4090), which carries the ledger read-only as a monitor rather than enforcing it, reaches 5.0–8.9 ms/step at 16×4 across four scenes (the two above, a 2.5×1.5 m road slab under a dropped crate, and the table). The

shelf and ledge scenes meet a 120 Hz budget there with margin (1.7× and 1.3×); the road scene sits at the boundary (1.01× on the mean, with a 9.3 ms worst step); the table scene does not (0.94×). All four meet it at 16×2 or below. We make no unqualified real-time claim. A device-resident *enforced* γ does not exist in this work, so governed device timings lie outside our measurements rather than being withheld (§4).

4 Limitations

Stability is not accuracy. Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory *away* from the converged reference even as it moves the energy toward it (§3.3). The result is bounded, not faithful; the honest use of the bound is as a safety envelope for budgets that would otherwise diverge. A natural next mechanism scales deviations about the quasi-static equilibrium rather than the whole state, preserving the load-bearing sag; we have not built it, and its unscaled equilibrium term would leave (2) holding only while the reservoir exceeds the sag’s strain energy.

Convergence remains the better remedy where affordable (§3.2).

Scope of the row. The coupling is normal-only — friction remains body-local, so scraping and rolling transfer lie outside it — and the position-level form fixes $e = 0$.

The bound governs modal storage only, not energy created spuriously in rigid degrees of freedom — in the impulse realization, box–box rows rectifying a support ring into net rigid motion. The ledger is blind to this by design.

The supply is a gross sum, so it can be recycled. Because the reservoir is funded by $\sum_k \max(\Delta E_{\text{rig}}^k, 0)$, energy returning from the modes to the rigid bodies and dissipated there is credited twice. We bound the exposure by accumulating the opposite channel, $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$, as a fraction of the supply: 0.4–27% (implicit) and 3–32% (position-based) — larger than the $\leq 1\%$ we expected, so the caveat rests on measurement. On the augmented-Lagrangian host it reaches 102–118%: the rigid subsystem’s gross *gain* exceeds its gross loss in every cell — the box–box rectification above, not a property of the bound. There the supply is an upper envelope on genuinely dissipated energy; tightening it would need the signed or per-interface accounting this scalar reservoir deliberately avoids.

Enforcement is host-side and single-machine (CPU float64); a device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work.

Formulation coverage. Three formulations, three scenes, one contact regime; the augmented-Lagrangian host’s small but near-universal overdraft makes any claim that descent-class solvers are passive in this setting an empirical statement about this sweep, not a theorem.

5 Conclusion

Transcribing an established rigid–modal contact law into fixed-budget solvers is not energy-safe, and how unsafe depends strongly on the formulation: worst-case modal-to-incident ratios are 1.2×10^5 , 1.70 and 0.53 across three hosts on the same 24 cells. The position-based amplification is a truncation artifact convergence removes; the augmented-Lagrangian overdraft we measure but do not explain.

Where convergence is unaffordable, a cumulative storage bound funded by measured gross rigid-side loss holds in every measured cell, at a quantified cost to contact accuracy.

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